

PCB Designer Handoff Generator

Generate a build-oriented 2-layer ESP32 sensor-board handoff from a controlled natural-language requirement.

Supported Scope Before You Enter a Requirement

Sensors available: temperature, humidity, light, air quality, pressure.

Fixed board architecture: USB-C 5 V power → 3.3 V regulator → ESP32-WROOM-32 → shared I2C bus.

What can change: the requested sensing functions and selected sensor footprints.

Out of scope examples: gps, camera, microphone, sound, audio, video, accelerometer, arbitrary chips, auto-routing, and manufacturing-ready Gerber generation.

➤ Sensor Plugin Validation

Board requirements

Make me a USB-C powered indoor monitoring board with WiFi, Bluetooth, temperature, humidity, and light sensing.

Refresh PCB Handoff

1. Requirement and Controlled Decisions

Project Summary

Item	Decision
Original requirement	Make me a USB-C powered indoor monitoring board with WiFi, Bluetooth, temperature, humidity, and light sensing.
Board type	2-layer FR-4 ESP32 sensor prototype
Power input	USB-C connector used for 5 V power only; no USB data routing
Regulated rail	Single 3.3 V rail for ESP32 and all sensors
Controller	ESP32-WROOM-32 module with WiFi/Bluetooth enabled
Sensors populated	AHT20, BH1750

Design Assumptions

Assumption	Value
PCB stackup	2-layer FR-4, 1.6 mm thickness, 1 oz copper, solder mask both sides
Assembly	Small-batch SMT assembly with hand-accessible test points
USB-C role	Sink-only power input; CC1/CC2 each use 5.1 kOhm pull-down to GND
Regulator	Use a 3.3 V LDO rated at 600 mA or higher with manufacturer-recommended capacitors
I2C bus	ESP32 GPIO21=SDA and GPIO22=SCL, 4.7 kOhm pullups to 3.3 V
Unsupported requests	None

Parsed Requirement JSON

```
▼ {
  "power_input" : "USB-C 5 V power only"
  ▼ "wireless" : [
    0 : "WiFi"
    1 : "Bluetooth"
  ]
  "mcu" : "ESP32-WROOM-32"
  "logic_voltage" : "3.3 V"
```

```
"communication_bus" : "I2C"  
▼ "requested_sensing" : [  
  0 : "temperature"  
  1 : "humidity"  
  2 : "light"  
]  
▼ "selected_components" : [  
  0 : "AHT20"  
  1 : "BH1750"  
]  
▶ "unsupported_requirements" : []  
}
```

2. Electrical Inputs and Outputs

I/O Definition

Interface	Direction	Pins/Nets	Electrical	Designer note
USB-C power input	Input	J1 VBUS to VBUS_5V, J1 GND to GND, CC1/CC2 to GND through 5.1 kOhm	5 V nominal from USB source	Do not route D+/D- unless a future USB data feature is added.
3.3 V system rail	Internal output	U1 VOUT to 3V3	3.3 V regulated, target ≥ 600 mA LDO	Feed ESP32, I2C pullups, sensors, status LED, and test point TP2.
Wireless	RF I/O	ESP32 module antenna	WiFi, Bluetooth	Keep copper, components, enclosure metal, and mounting hardware out of antenna keepout.
I2C sensor bus	Bidirectional	GPIO21/I2C_SDA, GPIO22/I2C_SCL	3.3 V logic with 4.7 kOhm pullups	Route as short, quiet traces away from regulator switching/noisy edges.
Status LED	Output	ESP32 GPIO2 to D1 through R5	Approx. 2-5 mA indicator current	Use for firmware heartbeat or fault indication.
Reset button	Input	ESP32 EN pulled up by R6, SW1 pulls EN to GND	Active-low reset	Place near board edge or accessible enclosure opening.
AHT20 sensor output	Input to MCU	AHT20 at I2C address 0x38 on I2C_SDA/I2C_SCL	3.3 V	Temperature and humidity
BH1750 sensor output	Input to MCU	BH1750 at I2C address 0x23 on I2C_SDA/I2C_SCL	3.3 V	Ambient light

3. Source-of-Truth Engineering Tables

Bill of Materials

Ref	Item	Value / Part	Qty	Footprint / Package	Role	Notes
J1	USB-C receptacle	USB-C power sink connector	1	USB-C 16-pin SMD, power-only pins used	Power input	Expose at board edge; connect shell to chassis/GND through chosen EMC strategy.
U1	3.3 V LDO regulator	AP2112K-3.3 or equivalent ≥600 mA LDO	1	SOT-23-5 or selected regulator footprint	Power regulation	Verify pinout against selected manufacturer before PCB release.
U2	ESP32-WROOM-32	ESP32-WROOM-32 module	1	Espressif module land pattern	MCU and wireless	Respect antenna keepout exactly from module datasheet.
R1, R2	USB-C CC pull-downs	5.1 kOhm, 1%, 0603	2	0603	USB-C sink detection	One resistor from CC1 to GND and one from CC2 to GND.
R3, R4	I2C pullups	4.7 kOhm, 1%, 0603	2	0603	I2C bus bias	Pull SDA and SCL to 3V3 near the ESP32 or bus center.
R5	LED current limit	330 Ohm, 0603	1	0603	Status LED	Adjust value for selected LED brightness target.
R6	EN pull-up	10 kOhm, 0603	1	0603	ESP32 reset	Pull ESP32 EN to 3V3.
D1	Status LED	Green LED, 0603	1	0603 LED	Visual status	Place visible from enclosure or board top.
SW1	Reset pushbutton	Momentary tactile switch	1	SMD tactile switch	Manual reset	Connect between EN and GND.
C1	Regulator input	10 uF, 10 V, X5R, 0805	1	0805 or regulator-	VBUS decoupling	Place close to U1 VBUS and GND.

	capacitor	X5R/X7R		recommended		VIN and GND.
C2	Regulator output capacitor	10 uF, 6.3 V, X5R/X7R	1	0805 or regulator-recommended	3V3 decoupling	Place close to U1 VOUT and GND.
C3	Regulator bypass capacitor	0.1 uF, 16 V, X7R	1	0603	High-frequency decoupling	Place close to U1.
C4, C5	ESP32 rail capacitors	10 uF and 0.1 uF ceramic	2	0805/0603	MCU decoupling	Place near ESP32 3V3 pins.
TP1	5 V test point	SMD test pad	1	1.0-1.5 mm test pad	Bring-up measurement	Label VBUS_5V on silkscreen.
TP2	3.3 V test point	SMD test pad	1	1.0-1.5 mm test pad	Bring-up measurement	Label 3V3 on silkscreen.
TP3	GND test point	SMD test pad	1	1.0-1.5 mm test pad	Bring-up measurement	Label GND on silkscreen.
TP4, TP5	I2C test points	SMD test pads for SDA/SCL	2	1.0 mm test pad	Firmware/debug probing	Place near ESP32 or sensor bus header area.
H1-H4	Mounting holes	M2.5 or M3 plated/non-plated holes	4	Mechanical holes	Mechanical mounting	Keep clear of antenna zone and USB connector shell.
U3	AHT20	AHT20 Temperature and humidity sensor	1	AHT20 LGA sensor package or verified breakout-compatible SMD footprint	Selected sensor	I2C address 0x38; Place at board edge with vent opening; keep away from regulator, ESP32, USB connector heat, and status LED.
C6	AHT20 local decoupling	0.1 uF, 16 V, X7R	1	0603	Sensor decoupling	Place within 2-5 mm of U3 supply and ground pins.
				BH1750FVI WSOF-		I2C address 0x23; Place near a board/window

U4	BH1750	BH1750 Ambient light sensor	1	6 or selected module- compatible SMD footprint	Selected sensor	edge with a clear optical path; do not shadow with tall parts or enclosure ribs.
C7	BH1750 local decoupling	0.1 uF, 16 V, X7R	1	0603	Sensor decoupling	Place within 2-5 mm of U4 supply and ground pins.

Pin Map

Ref	Pin	Net	Connection / Function	Type
J1	VBUS A4/A9/B4/B9	VBUS_5V	USB-C 5 V input to regulator VIN	Power
J1	GND A1/A12/B1/B12	GND	USB-C ground and shield reference	Ground
J1	CC1	USB_CC1	Connect to R1 then GND	Configuration
J1	CC2	USB_CC2	Connect to R2 then GND	Configuration
U1	VIN	VBUS_5V	Input from USB-C VBUS with C1 nearby	Power
U1	VOUT	3V3	Regulated rail with C2/C3 nearby	Power
U1	GND	GND	Solid ground return	Ground
U1	EN	VBUS_5V	Tie high for always-on regulator unless selected LDO differs	Enable
U2 ESP32	3V3	3V3	Main module supply; place C4/C5 nearby	Power
U2 ESP32	GND pins	GND	Connect to continuous ground pour with multiple vias	Ground
U2 ESP32	GPIO21	I2C_SDA	Shared I2C data line to sensors and R3 pullup	Digital I/O
U2 ESP32	GPIO22	I2C_SCL	Shared I2C clock line to sensors and R4 pullup	Digital I/O
U2 ESP32	GPIO2	STATUS_LED	Drive D1 through R5; avoid forcing boot-invalid state	Digital output
U2 ESP32	EN	ESP32_EN	Pulled to 3V3 by R6; SW1 pulls to GND for reset	Reset
R1	1 / 2	USB_CC1 / GND	5.1 kOhm USB-C sink pull-down	Resistor
R2	1 / 2	USB_CC2 / GND	5.1 kOhm USB-C sink pull-down	Resistor
R3	1 / 2	I2C_SDA / 3V3	4.7 kOhm pullup	Resistor
R4	1 / 2	I2C_SCL / 3V3	4.7 kOhm pullup	Resistor
D1/R5	Anode/Cathode path	3V3 / STATUS_LED	LED current path controlled by ESP32 GPIO2	Indicator
SW1	1 / 2	ESP32_EN / GND	Momentary reset button	Switch
TP1-	.	VBUS_5V, 3V3, GND,	Bring-up and firmware debug	

TP5	Pads	I2C_SDA, I2C_SCL	access	Test
U3	VDD	3V3	AHT20 Temperature and humidity sensor pin	Sensor
U3	GND	GND	AHT20 Temperature and humidity sensor pin	Sensor
U3	SDA	I2C_SDA	AHT20 Temperature and humidity sensor pin	Sensor
U3	SCL	I2C_SCL	AHT20 Temperature and humidity sensor pin	Sensor
U4	VCC	3V3	BH1750 Ambient light sensor pin	Sensor
U4	GND	GND	BH1750 Ambient light sensor pin	Sensor
U4	SDA	I2C_SDA	BH1750 Ambient light sensor pin	Sensor
U4	SCL	I2C_SCL	BH1750 Ambient light sensor pin	Sensor
U4	ADDR	GND	BH1750 Ambient light sensor pin	Sensor

PCB Netlist

Net	Connected pins	Type	Voltage	Routing / PCB note
VBUS_5V	J1 VBUS pins; U1 VIN; U1 EN; C1; TP1	Power	5 V	Route as wider power trace from USB-C to regulator; keep short.
3V3	U1 VOUT; U2 3V3; R3; R4; R6; C2-C5; TP2; sensor VDD/VCC pins	Power	3.3 V	Use a short rail or pour; decouple each IC locally.
GND	J1 GND; U1 GND; U2 GND; capacitors; R1; R2; SW1; TP3; all sensor GND pins	Ground	0 V	Use bottom-layer ground pour with stitching vias and uninterrupted sensor returns.
USB_CC1	J1 CC1; R1	USB-C configuration	0-5 V	Short trace; R1 returns to GND.
USB_CC2	J1 CC2; R2	USB-C configuration	0-5 V	Short trace; R2 returns to GND.
I2C_SDA	U2 GPIO21; R3; TP4; sensor SDA pins	Digital bus	3.3 V	Route as shared bus with short stubs; avoid antenna area.
I2C_SCL	U2 GPIO22; R4; TP5; sensor SCL pins	Digital bus	3.3 V	Route alongside SDA where practical; keep away from noisy power loop.
STATUS_LED	U2 GPIO2; R5; D1	Digital output	3.3 V	Confirm GPIO2 boot behavior with firmware and LED polarity.
ESP32_EN	U2 EN; R6; SW1	Reset	3.3 V	Keep trace short and away from noisy nets; optional 0.1 uF EN capacitor may be added if reset is unstable.
AHT20_LOCAL_DECOUPLING	U3 supply pin; local 0.1 uF capacitor; GND	Decoupling	3.3 V	Place capacitor next to U3; Short I2C stubs; no copper pour directly under sensing opening

	GND			if using exposed environmental port.
BH1750_LOCAL_DECOUPLING	U4 supply pin; local 0.1 uF capacitor; GND	Decoupling	3.3 V	Place capacitor next to U4; Keep light sensor area free from tall silkscreen labels, LEDs, and dark solder-mask obstructions.

Power Budget

Block	Estimated current	Rail	Notes
ESP32-WROOM-32	300 mA peak	3V3	Budget for WiFi/Bluetooth transmit peaks, not just idle current.
Status LED	5 mA	3V3	Depends on R5 and selected LED brightness.
I2C pullups	1.4 mA worst case	3V3	Approx. two 4.7 kOhm pullups when both lines are low.
AHT20	1.0 mA	3V3	Temperature and humidity sensor at I2C address 0x38.
BH1750	0.2 mA	3V3	Ambient light sensor at I2C address 0x23.
3.3 V regulator requirement	307.6 mA estimated peak load	3V3	Use >=600 mA LDO; estimated margin is 292.4 mA before thermal derating.
Thermal check	Review during layout	VBUS_5V to 3V3	LDO heat is roughly (5 V - 3.3 V) * load current; provide copper area near U1.

4. Schematic and PCB Implementation Notes

Schematic Summary

- J1 receives USB-C 5 V power. VBUS feeds U1 VIN and GND ties to the board ground system.
- R1 and R2 are 5.1 kOhm pull-down resistors from CC1 and CC2 to GND so a USB-C source enables 5 V.
- U1 regulates VBUS_5V to 3V3. C1 is the input bulk capacitor; C2/C3 are output and high-frequency bypass capacitors.

- U2 ESP32-WROOM-32 is powered from 3V3. GPIO21 is I2C_SDA and GPIO22 is I2C_SCL.
- R3 and R4 pull I2C_SDA and I2C_SCL to 3V3. TP4 and TP5 expose the bus for debug.
- D1/R5 provide a status LED from GPIO2, and SW1 pulls ESP32_EN low for reset while R6 pulls EN high.
- U3 AHT20 connects to 3V3, GND, I2C_SDA, and I2C_SCL at address 0x38; add local 0.1 uF decoupling.
- U4 BH1750 connects to 3V3, GND, I2C_SDA, and I2C_SCL at address 0x23; add local 0.1 uF decoupling.

2-Layer PCB Layout Instructions

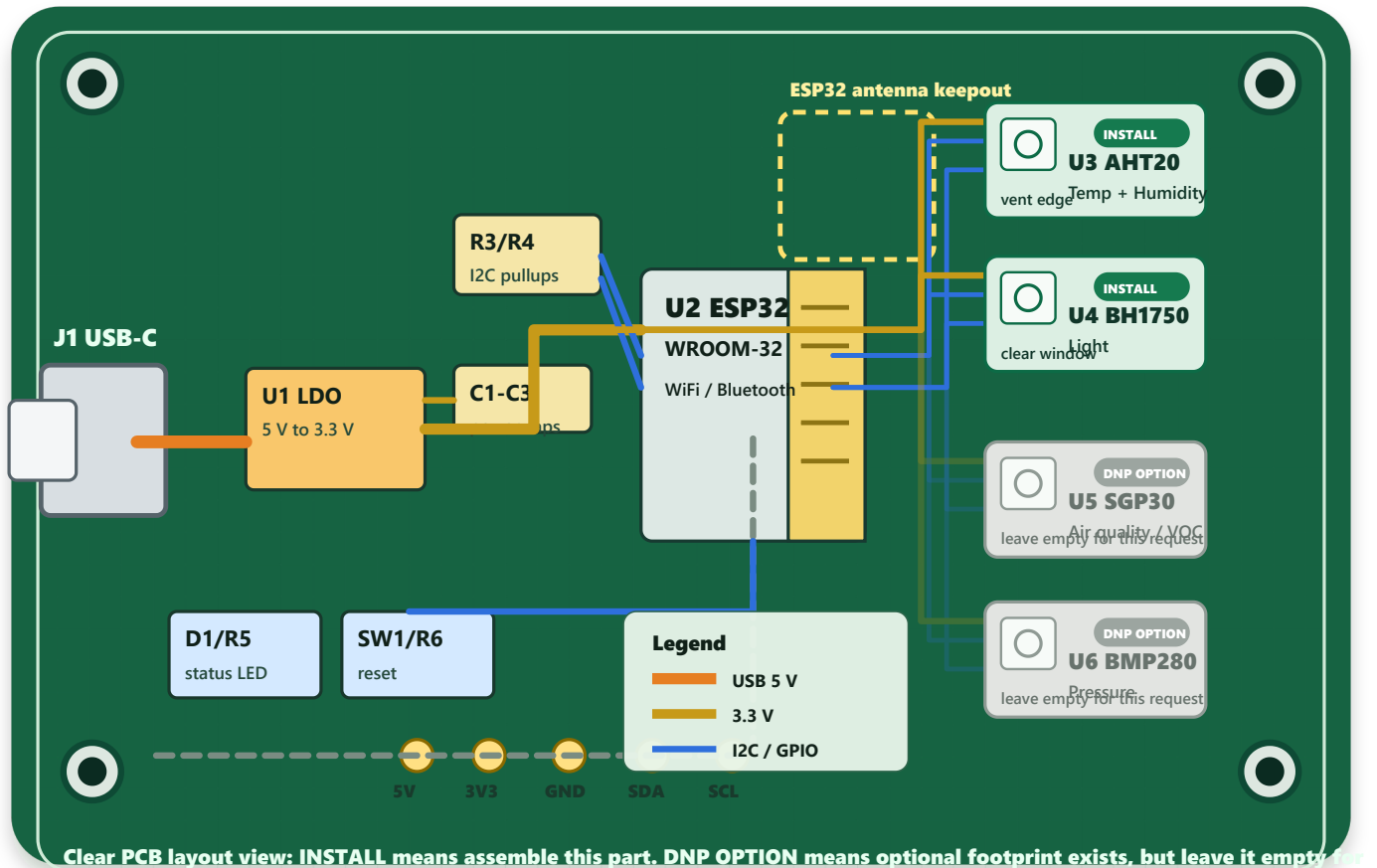
Area	Instruction
Board outline	Target a compact rectangle around 45 mm x 35 mm unless enclosure forces another shape; place mounting holes near corners.
USB-C edge	Place J1 centered on one board edge with connector overhang and shell pads per footprint drawing.
Power zone	Place U1, C1, C2, and C3 within 10 mm of J1; keep VBUS_5V trace short and wider than signal traces.
ESP32 module	Place U2 with its antenna at the opposite board edge or hanging over a board edge; keep the full antenna keepout free of copper on all layers.
Grounding	Use bottom layer as continuous GND pour; stitch top/bottom GND near USB shell, regulator, ESP32, and sensors.
I2C bus	Route SDA/SCL from ESP32 to sensors as short 3.3 V digital traces with R3/R4 near ESP32 or bus midpoint; add TP4/TP5 before branches.
Decoupling	Every IC gets local 0.1 uF decoupling from supply to GND with the shortest possible return path.
User access	Place D1 and SW1 on the top side where visible/pressable after assembly or through enclosure openings.
Test access	Place TP1 VBUS_5V, TP2 3V3, TP3 GND, TP4 SDA, and TP5 SCL along one accessible board edge.
U3 AHT20	Place at board edge with vent opening; keep away from regulator, ESP32, USB connector heat, and status LED.
U4 BH1750	Place near a board/window edge with a clear optical path; do not shadow with tall parts or enclosure ribs.

5. Visual Diagrams

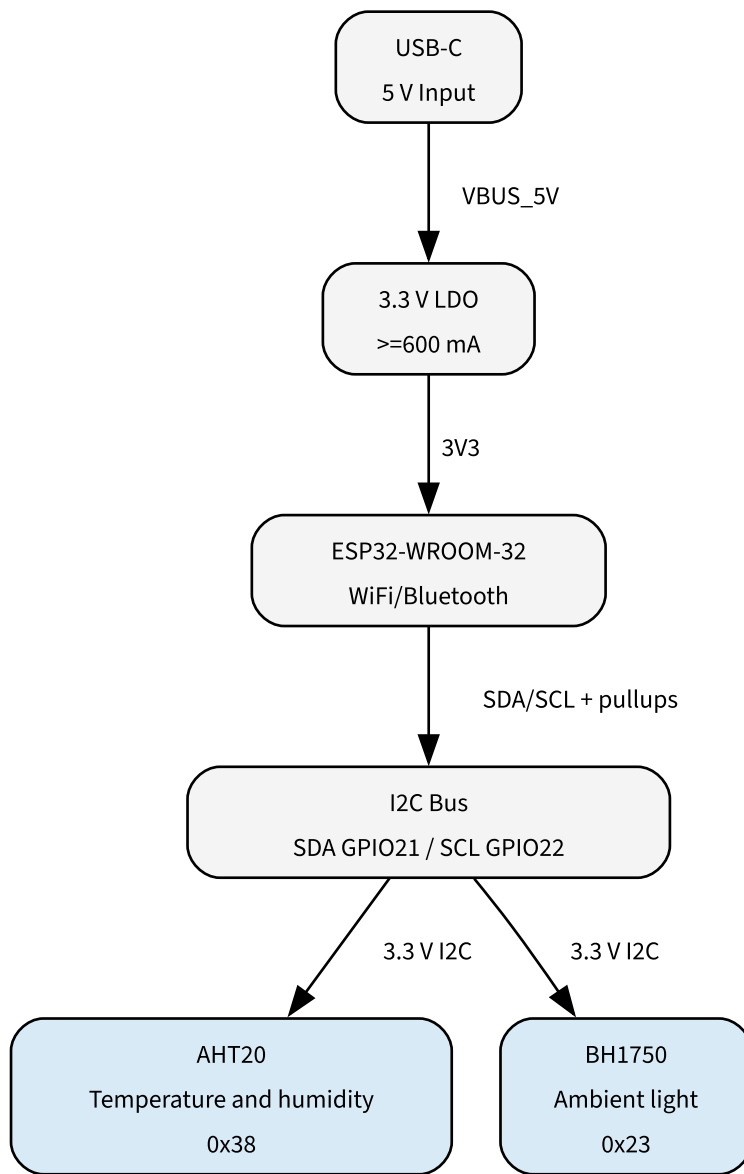
Diagrams are implementation aids. The tables above are the source of truth for PCB capture.

PCB Layout Visual

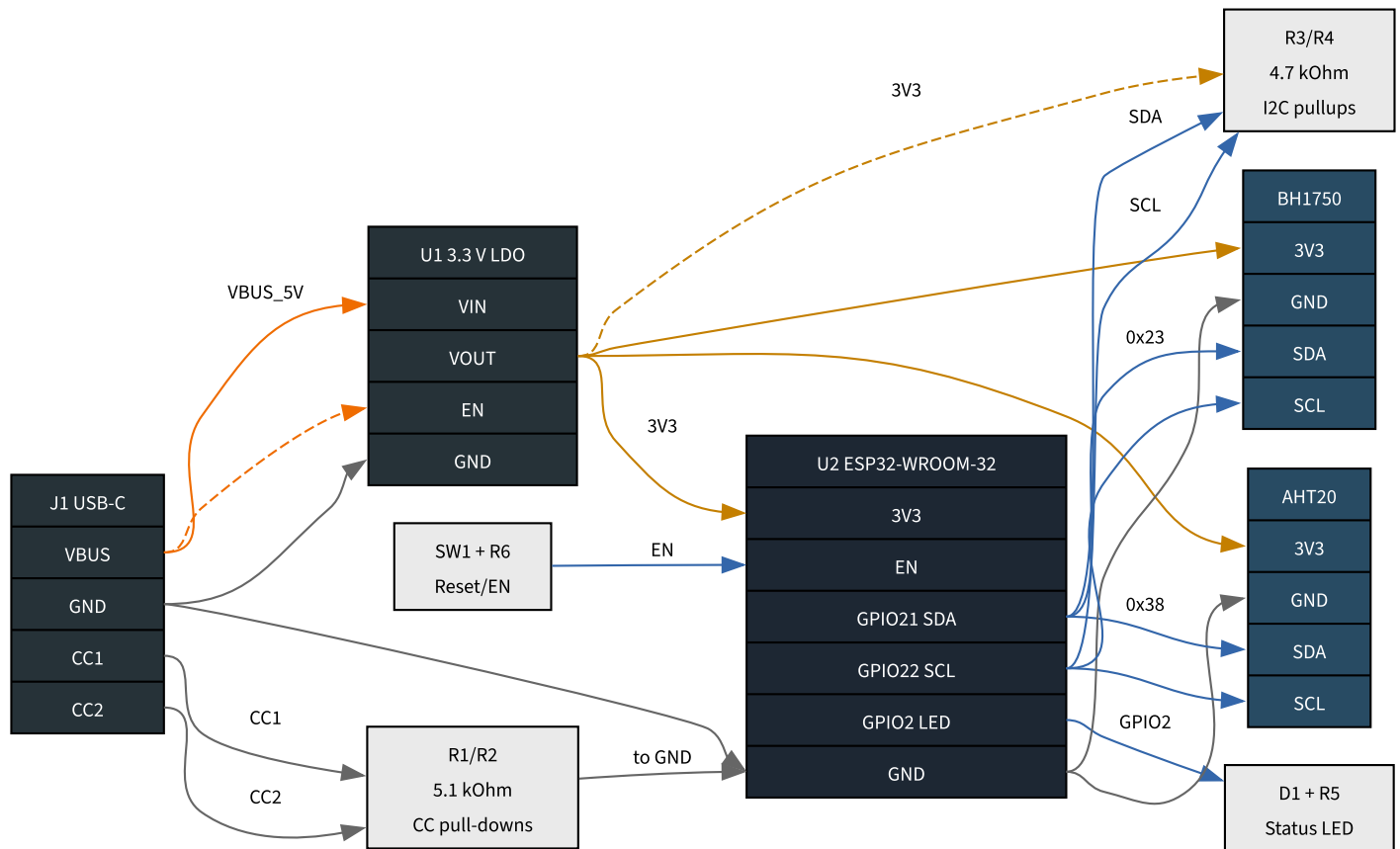
How to read this board: sensors marked **INSTALL** are part of the current design. Sensors marked **DNP OPTION** are optional footprints on the shared prototype PCB; DNP means 'do not populate,' so those parts are left empty during assembly for this user request.



System Architecture



Schematic Connectivity



6. Build Checks

Fabrication and Assembly Checklist

- **DRC:** No clearance, unconnected net, solder-mask, or silkscreen-on-pad errors.
- **Antenna keepout:** ESP32 antenna area has no copper, traces, vias, components, or mounting hardware on any layer.
- **USB-C footprint:** Connector orientation, shell pads, VBUS/GND pins, and CC pins match the purchased part.
- **Regulator footprint:** U1 pinout matches selected LDO; capacitors meet datasheet ESR/capacitance requirements.
- **Sensor exposure:** Environmental sensors are not blocked by enclosure, solder mask openings, adhesive, coating, or tall parts.
- **Silkscreen:** Reference designators, polarity, USB orientation, and test point labels are readable.

- **Assembly notes:** Mark SGP30/environmental sensors as no-wash/no-conformal-coat if populated.

Bring-Up Checklist

- **Pre-power inspection:** No solder bridges on USB-C, ESP32 module pads, regulator, or sensor pins.
- **Resistance check:** No short between VBUS_5V and GND or 3V3 and GND before connecting USB power.
- **Apply USB power:** TP1 measures about 5 V and TP2 measures 3.3 V.
- **Reset check:** Pressing SW1 pulls ESP32_EN low and releases back to 3.3 V.
- **Firmware flash:** ESP32 boots and toggles D1 status LED.
- **I2C scan:** Firmware detects every populated sensor address.
- **Verify U3 AHT20:** I2C address 0x38 responds and reports plausible temperature and humidity data.
- **Verify U4 BH1750:** I2C address 0x23 responds and reports plausible ambient light data.